

CS & IT ENGINEERING



Computer Network

Flow Control

Lecture No. - 09



By - Abhishek Sir



Recap of Previous Lecture



Topic

Selective Repeat ARQ ✓





Topics to be Covered



Topic

End-to-End Delay

Topic

Optimal Packet Size

ABOUT ME



Hello, I'm **Abhishek**

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#Q. Consider a 128×10^3 bits/second satellite communication link with one-way propagation delay of 150 milliseconds. Selective retransmission (repeat) protocol is used on this link to send data with a frame size of 1 kilobyte. Neglect the transmission time of acknowledgment. The minimum number of bits required for the sequence number field to achieve 100% utilization is _____.

[GATE-2016, Set-2, 2-Mark]

Solution:-

$$\text{Packet Size} = 1 \text{ KB} = 2^{13} \text{ bits}$$

$$\text{Bandwidth} = 128 * 10^3 \text{ bits / sec}$$

$$t_x = \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{2^{13} \text{ bits}}{2^7 * 10^3 \text{ bits / sec}} = 64 \text{ ms}$$

$$t_p = 150 \text{ ms} \checkmark$$

$$\begin{aligned} \text{Cycle time} &= (t_x + 2 * t_p) = 364 \text{ ms} \\ &= (64 + 2 * 150) \text{ ms} \end{aligned}$$

$$\text{Optimal Window Size (N)} = \left\lceil \frac{\text{Cycle Time (RTT)}}{\text{Transmission delay}} \right\rceil = \left\lceil \frac{364 \text{ ms}}{64 \text{ ms}} \right\rceil = 6$$

For Selective Repeat ARQ :

$$\text{Total number of sequences} = (N + N) = 2 * N = 12$$

Minimum number of bits required for sequence number field

$$\begin{aligned} &= \lceil \log_2 [\text{Total number of sequences}] \rceil \text{ bits} \\ &= \lceil \log_2(12) \rceil \text{ bits} \\ &= 4 \text{ bits} \end{aligned}$$

$$\boxed{\text{Ans} = 4}$$

#Q. Consider a selective repeat sliding window protocol that uses a frame size of 1 KB to send data on a 1.5 Mbps link with a one-way latency of 50 msec. To achieve a link utilization of 60%, the minimum number of bits required to represent the sequence number field is _____.

[GATE-2014, Set-1, 2-Mark]

Solution:-

$$\text{Packet Size} = 1 \text{ KB} = 2^{13} \text{ bits}$$

$$\text{Bandwidth} = 1.5 \text{ Mbps} = 1.5 * 10^6 \text{ bits / sec}$$

$$t_x = \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{2^{13} \text{ bits}}{1.5 * 10^6 \text{ bits / sec}} = 5.461 \text{ ms}$$

$$t_p = 50 \text{ ms}$$

$$\text{Cycle time} = (t_x + 2 * t_p) = 105.461 \text{ ms}$$

To achieve 60% utilization ($\eta = 3/5$) in ~~Go Back 127 ARQ~~ sliding window (selective Repeat)

$$\text{Efficiency } (\eta) = \frac{N * \text{Transmission delay}}{\text{Cycle Time}}$$

$$N = \left\lceil \frac{3 * \text{Cycle Time}}{5 * \text{Transmission delay}} \right\rceil$$

$$= \left\lceil \frac{3 * 105.461 \text{ ms}}{5 * 5.461 \text{ ms}} \right\rceil$$

$$N = 12$$

For Selective Repeat ARQ:

$$\text{Total number of sequences} = \underbrace{(N + N)} = \boxed{2 * N} = 24$$

Minimum number of bits required for sequence number field

$$= \lceil \log_2 [\text{Total number of sequences}] \rceil \text{ bits}$$

$$= \lceil \log_2(24) \rceil \text{ bits}$$

$$= 5 \text{ bits}$$

$$\boxed{\text{Ans} = 5}$$

#Q. In a sliding window ARQ scheme, the transmitter's window size is N and the receiver's window size is M . The minimum number of distinct sequence numbers required to ensure correct operation of the ARQ scheme is

[GATE 2004]

A $\min(M, N)$

B $\max(M, N)$

☒ **C** $M + N$

D MN

Ans: C

Solution :-

Transmitter's transmitting window size = N

Receiver's receiving window size = M

Total number of sequences = $(N + M)$

Total number of sequences =
 Transmitter's transmitting window size
 + Receiver's receiving window size

$(N > 1)$

For Go Back N ARQ
 $M = 1$

For selective Repeat ARQ
 $M = N$

Transmitter's transmitting window size = N

1. Sliding Window Protocol

Total number of sequences = N

2. Go Back N ARQ

Total number of sequences = $(N+1)$

3. Selective Repeat ARQ

Total number of sequences = $(N+N) = (2N)$

Minimum number of bits required for sequence number field

$$= \lceil \log_2(\text{Total number of sequences}) \rceil \text{ bits}$$

Number of bits in sequence number field = k bits

Total number of sequences = 2^k

$$0 \text{ to } (2^k - 1)$$

Number of bits in sequence number field = k bits

1. Sliding Window Protocol

Transmitter's transmitting window size = 2^k

2. Go Back N ARQ

Transmitter's transmitting window size = $(2^k - 1)$

3. Selective Repeat ARQ

Transmitter's transmitting window size = $2^{(k-1)}$

#Q. The maximum window size for data transmission using the selective reject ^{Repeat.} protocol with n-bit frame sequence numbers is:

[GATE 2005]

IIT-B

A $2^n \Rightarrow$ sliding window

✓ **B** $2^{(n-1)} \Rightarrow$ selective Repeat

C $(2^n) - 1 \Rightarrow$ Go Back N ARQ

D $2^{(n-2)}$

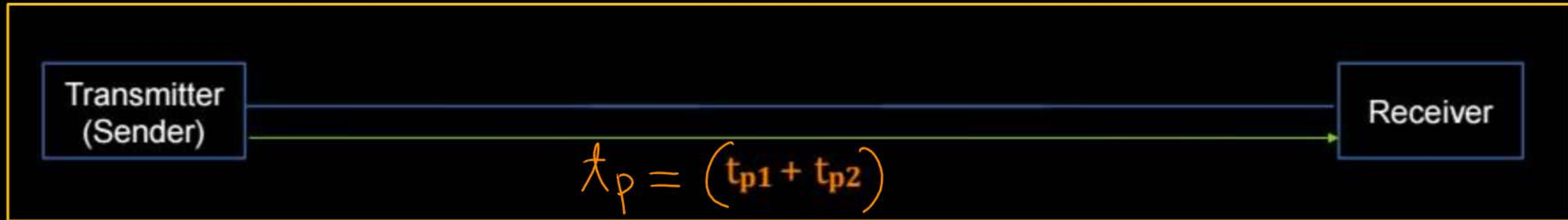
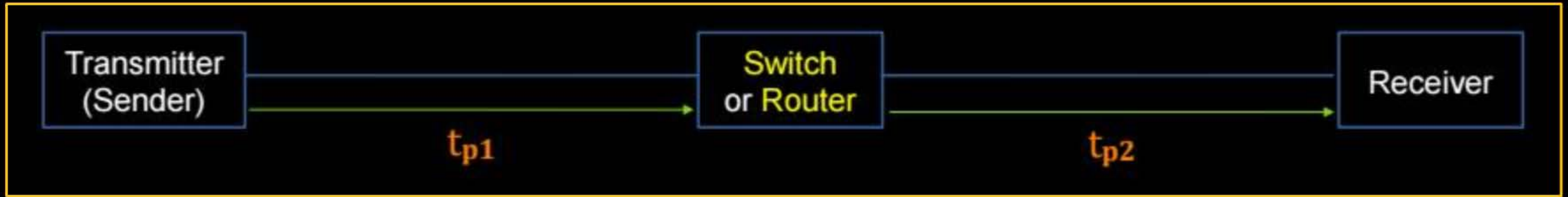
Ans: B



Topic : Store and Forward Device

→ Layer-2 or Layer-3 device

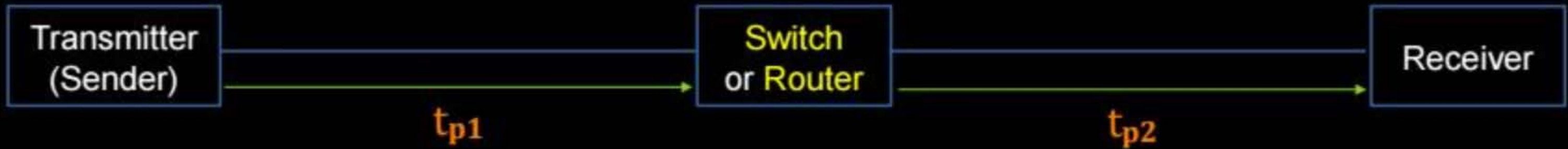
→ Switch or Router





Topic : Store and Forward Device

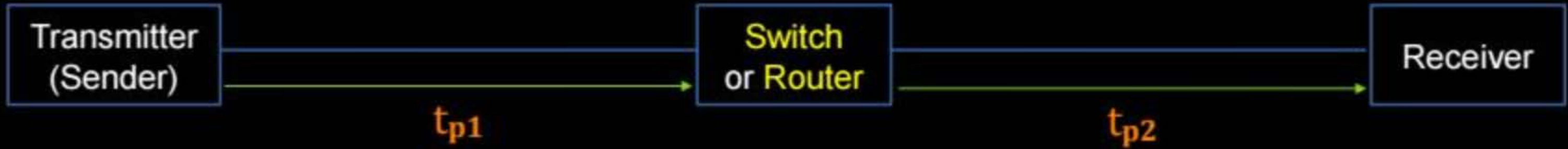
→ Store and Forward device
[Store, Process and Forward]



$$\text{Store and Forward delay} = \text{Queuing delay at device} + \text{Processing delay by device} + [\text{Transmission delay}]$$



Topic : End-to-End Delay



→ Suppose each link have same bandwidth and same propagation delay

→ Time required for a packet to be transmitted from Transmitter to Receiver

$$\begin{aligned} \text{End-to-end delay (one-way)} &= \text{Transmission delay}_1 + \text{Propagation delay}_1 \\ &+ [\text{Queuing delay at device} + \text{Processing delay by device} \\ &+ \text{Transmission delay}_2] + \text{Propagation delay}_2 \end{aligned}$$

$$= (t_x + t_{p1}) + \text{Processing Delay} + (t_x + t_{p2})$$

Example 9 :-

Consider two hosts A and B are connected through one router (router takes negligible processing delay). Suppose each link bandwidth is 1 Kbps and each link propagation delay is 200 milliseconds. Calculate amount of time required (in milliseconds) to transmit one 10 byte packet from hosts A to B?

$$t_x = \frac{\text{Packet size}}{\text{Bandwidth}} = \frac{10 \text{ bytes}}{1 \text{ Kbps}} = 80 \text{ ms} \quad | \quad t_{p_1} = t_{p_2} = 200 \text{ ms}$$

① Without Router $\text{Total Time} = [t_x + (t_{p_1} + t_{p_2})] = 480 \text{ ms}$

② With Router $\text{Total Time} = (t_x + t_{p_1}) + \text{Processing Delay} + (t_x + t_{p_2})$
 $= (80 + 200) + 0 + (80 + 200) \text{ ms}$
 $= 560 \text{ ms}$

Ans = 560

Solution:-

$$\text{Packet Size} = 10 \text{ bytes} = 80 \text{ bits}$$

$$\text{Bandwidth} = 1 \text{ Kbps} = 10^3 \text{ bits / sec}$$

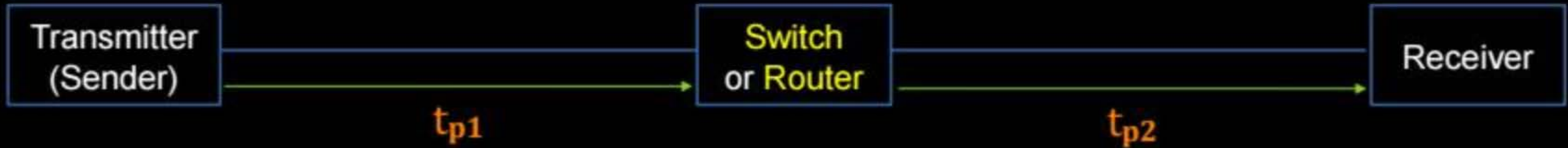
$$t_x = \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{80 \text{ bits}}{10^3 \text{ bits / sec}} = 80 \text{ ms}$$

$$t_p = 200 \text{ ms}$$

$$\begin{aligned} \text{End-to-end delay} &= (t_x + t_p) + (t_x + t_p) \\ &= 560 \text{ ms} \end{aligned}$$

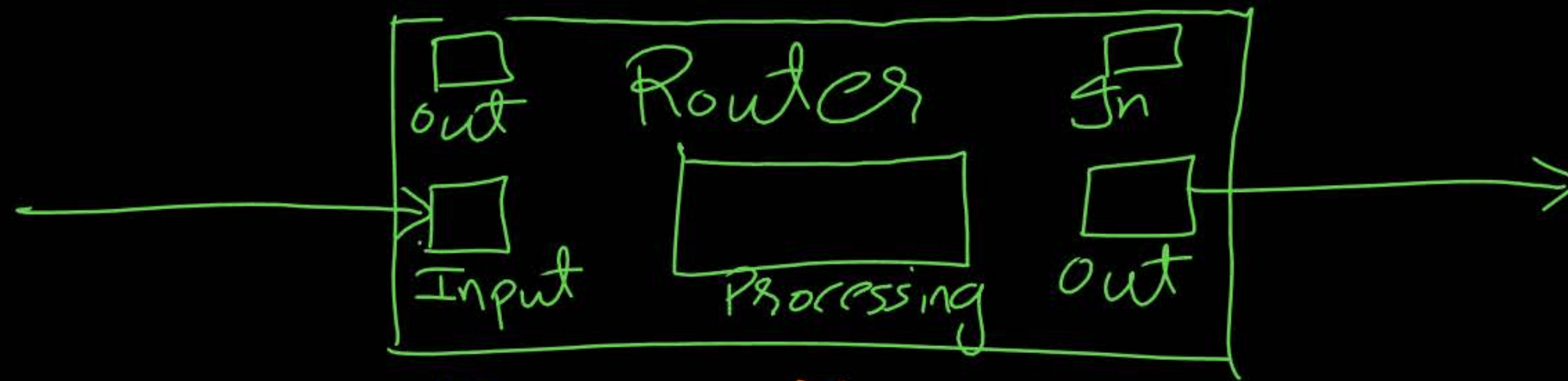


Topic : End-to-End Delay



- Suppose each link have same bandwidth and same propagation delay
- Time required for N packet to be transmitted from Transmitter to Receiver

$$\begin{aligned} \text{End-to-end delay} = & \boxed{(N * \text{Transmission delay})} + \text{Propagation delay}_1 \\ & + [\text{Queuing delay at device} + \text{Processing delay by device} \\ & + \text{Transmission delay}] + \text{Propagation delay}_2 \end{aligned}$$


 F_3
 F_2
 F_1
 F_4
 F_3
 F_2
 F_N
 $F_{(N-1)}$
 $F_{(N-2)}$
 F_N
 $F_{(N-1)}$
 F_N

Total Time (End-to-End Delay)

$$= (N \times t_x + t_{P_1}) + \text{Processing Delay} + (t_x + t_{P_2})$$

Example 10 :-

Consider two hosts A and B are connected through one router (router takes negligible processing delay). Suppose packet size is 20 bytes, each link bandwidth is 2 Mbps and each link propagation delay is 500 microseconds. Calculate amount of time required (in microseconds) to transmit a file of 100 bytes from hosts A to B?

① Without Router
$$\text{Total Time} = N \times t_x + (t_{p1} + t_{p2})$$
$$= 1400 \mu s$$

② With Router
$$\text{Total Time} = 1480 \mu s$$

$$\text{Ans} = 1480$$

Solution:-

$$\text{Packet Size} = 20 \text{ bytes} = 160 \text{ bits}$$

$$\text{Bandwidth} = 2 \text{ Mbps} = 2 * 10^6 \text{ bits / sec}$$

$$t_x = \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{160 \text{ bits}}{2 * 10^6 \text{ bits / sec}} = 80 \mu\text{s}$$

~~$$t_p = 500 \mu\text{s}$$~~
$$t_{p1} = t_{p2} = 500 \mu\text{s}$$

$$\text{File Size} = 100 \text{ bytes}$$

$$\text{Packet Size} = 20 \text{ bytes}$$

$$\text{Number of packets (N)} = \frac{\text{File Size}}{\text{Packet Size}} = \frac{100 \text{ bytes}}{20 \text{ bytes}} = 5$$

$$\begin{aligned} \text{End-to-end delay} &= (N * t_x + t_p) + (t_x + t_p) = (5 * 80 + 500) + (80 + 500) \mu s \\ &= 1480 \mu s \end{aligned}$$

#Q. Consider a network path P—Q—R between nodes P and R via router Q. Node P sends a file of size 10^6 bytes to R via this path by splitting the file into chunks of 10^3 bytes each. Node P sends these chunks one after the other without any wait time between the successive chunk transmissions. Assume that the size of extra headers added to these chunks is negligible, and that the chunk size is less than the MTU.

Each of the links P—Q and Q—R has a bandwidth of 10^6 bits/sec, and negligible propagation latency. Router Q immediately transmits every packet it receives from P to R, with negligible processing and queueing delays. Router Q can simultaneously receive on link P—Q and transmit on link Q—R.

Assume P starts transmitting the chunks at time $t = 0$. Which one of the following options gives the time (in seconds, rounded off to 3 decimal places) at which R receives all the chunks of the file?

(A) 8.000

(B) 8.008

(C) 15.992

(D) 16.000

[GATE 2024]

11SC / H.W.

#Q. Two hosts are connected via a packet switch with 10^7 bits per second links. Each link has a propagation delay of 20 microseconds. The switch begins forwarding a packet 35 microseconds after it receives the same. If 10000 bits of data are to be transmitted between the two hosts using a packet size of 5000 bits, the time elapsed between the transmission of the first bit of data and the reception of the last bit of the data in microseconds is _____.

[GATE 2015]

11T-K, H.W.

#Q. Consider a source computer (S) transmitting a file of size 10^6 bits to a destination computer (D) over a network of two routers (R1 and R2) and three links (L1, L2, and L3). L1 connects S to R1; L2 connects R1 to R2; and L3 connects R2 to D. Let each link be of length 100 km. Assume signals travel over each link at a speed of 10^8 meters per second. Assume that the link bandwidth on each link is 1Mbps. Let the file be broken down into 1000 packets each of size 1000 bits. Find the total sum of transmission and propagation delays in transmitting the file from S to D.

- (A) 1005 ms
- (B) 1010 ms
- (C) 3000 ms
- (D) 3003 ms

[GATE 2012]
 IIT-D, H.W.



Topic : Optimal Packet Size



→ Suppose each link have same bandwidth

$$\underline{t_x} = \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{(\underline{H} + \underline{P})}{\text{Bandwidth}}$$

K = Number of Links = 3

M = Message Size

H = Packet Header Size

P = Packet Payload Size

$$\underline{\text{Number of packets (N)}} = \frac{\text{Message Size}}{\text{Payload Size}} = \frac{M}{P}$$



Topic : Optimal Packet Size



→ Total time required to transmit one message completely from Sender to Receiver

$$\text{Total Time (T)} = [N * t_x + t_{p1}] + [t_x + t_{p2}] + [t_x + t_{p3}]$$

$$= (N + K - 1) * t_x + [t_{p1} + t_{p2} + t_{p3}]$$

$$= (N + K - 1) * t_x + t_p$$

[Consider negligible propagation delay] $[t_p = 0]$

$$= (N + K - 1) * t_x$$

$$T = \frac{(N + K - 1) * (H + P)}{\text{Bandwidth}}$$



Topic : Optimal Packet Size



$$\text{Total Time (T)} = (N + K - 1) * t_x$$

$$T = \frac{(N + K - 1) * (H + P)}{\text{Bandwidth}}$$

→ For what value of P, total time T should be minimum :

$$\left| \frac{dT}{dP} = 0 \right|$$

$$P = \sqrt{\frac{MH}{(K-1)}}$$

$$\text{Optimal packet size} = (H + P)$$

$$T = \frac{\left(\frac{M}{P} + K - 1\right) * (H + P)}{\text{Bandwidth} \cdot h}$$

$$\frac{dT}{dP} = \left(\frac{1}{\text{Bandwidth} \cdot h}\right) \frac{d}{dP} \left[\left(\frac{M}{P} + K - 1\right) * (H + P) \right]$$

$$\left(\frac{1}{\text{Band}}\right) \left[\left(\frac{M}{P} + K - 1\right) * \frac{d}{dP} (H + P) + (H + P) * \frac{d}{dP} \left(\frac{M}{P} + K - 1\right) \right] = 0$$

$$\left(\frac{M}{P} + K - 1\right) * 1 + (H + P) * \left(-\frac{M}{P^2}\right) = 0$$

$$\left(\frac{M + P * (K - 1)}{P}\right) = \frac{MH + PM}{P^2}$$

$$PM + P^2 * (K - 1) = MH + PM$$

$$P^2 = \frac{MH}{(K - 1)} \quad \bigg| \quad P = \sqrt{\frac{MH}{(K - 1)}}$$

#Q. In a packet switching network, packets are routed from source to destination along a single path having two intermediate nodes. If the message size is 24 bytes and each packet contains a header of 3 bytes, then the optimum packet size is:

$$(K-1) = \text{No. of nodes/Routers} = 2$$

$$K = \text{No. of links} = 3$$

[GATE 2005]

(IIT-B)

(A) 4

~~(B) 6~~

(C) 7

✓ (D) 9

$$\text{Message Size (M)} = 24 \text{ bytes}$$

$$\text{Header Size (H)} = 3 \text{ bytes}$$

$$\text{Optimum payload size (P)} = ?$$

$$\begin{aligned} \text{Optimum Packet Size} &= (H + P) \\ &= (3 + 6) \text{ bytes} \\ &= 9 \text{ bytes} \end{aligned}$$

$$P = \sqrt{\frac{MH}{(K-1)}} =$$

$$P = \sqrt{\frac{24 \text{ byte} \times 3 \text{ byte}}{2}}$$

$$P = 6 \text{ byte}$$

Ans: D



2 mins Summary



Topic

End-to-End Delay ✓

Topic

Optimal Packet Size ✓

(GATE 2014)

Framing:-

- ① Byte Count
- ② Byte stuffing
- ③ Bit stuffing



THANK - YOU

